

# Agentic metasurface design with self-correcting language-model systems

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## Abstract

Automated metasurface design is increasingly important, and recent advances in language-model systems are opening a route toward agentic optical design. Yet modern metasurface applications, from metalenses and holography to optical computing, require long design chains spanning modeling, simulation, coding, optimization and evaluation. These chains are error-prone, whereas existing language-model-based metasurface tools remain largely limited to simple objectives, predefined pipelines or language-to-layout generation. Here we introduce MetaDesigner, a self-correcting language-model system for agentic metasurface design. From a natural-language optical objective, MetaDesigner plans the design route, retrieves domain knowledge, invokes simulation and optimization tools, generates missing tool code and identifies errors through a dedicated Verifier. We demonstrate three tasks of increasing complexity: an RGB metalens with three independent focal spots, a six-plane full-color hologram with an average structural similarity index measure (SSIM) of 0.97, and an optoelectronic hybrid neural network for image style transfer. These tasks require 74, 136 and 90 reasoning steps, respectively, and the system self-corrects errors in frequency mapping, numerical aperture estimation, network-parameter counting and loss-function description. These results establish MetaDesigner as a self-correcting route to agentic metasurface design, where language-model systems can not only execute optical design tasks but also extend, inspect and repair the design process itself.

## Introduction

Optical metasurfaces<sup>1-3</sup> have reshaped optical design by replacing bulky refractive and diffractive components<sup>4</sup> with ultrathin arrays of engineered nanostructures. By tailoring phase<sup>5,6</sup>, polarization<sup>7,8</sup>, amplitude and momentum<sup>9,10</sup> at the subwavelength scale, they have enabled compact beam shapers<sup>11</sup>, metalenses<sup>12,13</sup>, holograms<sup>14-16</sup> and diffractive computing systems<sup>17-19</sup>. This versatility arises from an exceptionally large design space, in which material choice, unit-cell geometry and spatial arrangement can all serve as degrees of freedom. Over the past decade, adjoint optimization, topology optimization<sup>20</sup>, evolutionary search<sup>21</sup> and deep learning<sup>22,23</sup> have made parts of this space increasingly searchable. Yet these methods typically operate only after the essential design problem has been formulated by human experts. The designer must still decide which physics to model, which variables to optimize, which objective to impose, which solver to use and how to interpret the result. Thus, although modern inverse design has made metasurface optimization increasingly automated, the construction of the design process itself remains largely human-driven.

Language-model systems offer a route from automated optimization toward agentic metasurface design. By combining natural-language reasoning, domain-knowledge retrieval, code generation, tool use and iterative revision, they can in principle operate at the level where an optical intention is converted into an executable design process<sup>24-29</sup>. This level is particularly important for advanced metasurfaces. A functional device is not produced by optimization alone, but by a coupled chain of decisions involving physical modeling, electromagnetic simulation, data preparation, numerical propagation, loss-function design, code implementation, performance evaluation and technical interpretation. Recent agentic frameworks have begun to show that language-described photonic goals can be converted into device layouts<sup>30,31</sup>. However, as optical tasks become more complex, this chain becomes increasingly fragile. A wrong wavelength convention, an inappropriate propagation model, a hidden coding error or an overinterpreted metric can pass through optimization and yield a result that appears plausible but is physically unreliable. Agentic metasurface design therefore requires more than the ability to execute tasks: the system must also be able to inspect, verify and repair its own reasoning and computation.

Here we introduce MetaDesigner, a self-correcting language-model system for agentic metasurface design. Unlike previous metasurface agents that mainly translate language-described goals into device layouts within predefined pipelines, MetaDesigner plans, executes and verifies the design process itself. Starting from a natural-language optical objective, it decomposes the task into